MILITARY SPECIFICATION

EXPLOSIVE, PLASTIC-BONDED, CAST PBXN-109

This amendment forms a part of MIL-E-82886(OS) dated 18 June 1993, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

PAGE 16

6.6: Delete and substitute:

" **6.6 IDPI weight percent calculation.** The following method is used to calculate the weight percent of IDPI required to react with the following materials:

a. Method

Step 1: Calculate the IDPI required to react with PolyBD (OH-terminated):

$$X_1 = \frac{(Eq)\ (Y)\ (R)}{Ep}$$

where:
- X_1 = Weight percent of IDPI required to react with PolyBD
- Eq = Equivalent weight of IDPI (g/eq) (111.60 g/eq nominal, actual equivalent weight should be analyzed)
- Y = Weight percent of PolyBD (OH-terminated) = 7.346
- R = NCO/OH ratio (1.0/1 nominal)
- Ep = Equivalent weight of PolyBD (g/eq) (1204.82 g/eq nominal, actual equivalent weight must be analyzed)

Step 2: Calculate the IDPI required to react with DHE:

$$X_2 = \frac{(Eq)\ (Ohp)\ (Z)\ (R)}{(56.1\ mg\ KOH/eq)\ (1000\ mg/g)}$$

where:
- X_2 = Weight percent of IDPI required to react with DHE
- Eq = Equivalent weight of IDPI (g/eq) (111.60 g/eq nominal, actual equivalent weight should be analyzed)
- Ohp = Hydroxyl number of DHE (mg KOH/g) (500 mg KOH/g nominal, actual equivalent weight should be analyzed)

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Z = Weight percent of DHE = 0.26
R = NCO/OH ratio (1.0/1 nominal)

Step 3: Calculate the total IDPI weight percent:

$$T = X_1 + X_2$$

where: T = Total IDPI weight percent

b. Sample calculation (with nominal equivalent weights)

$$X_1 = \frac{(111.60\ g/eq)\ (7.346)\ (1)}{1204.82\ g/eq)} = 0.6804$$

$$X_2 = \frac{(111.60\ g/eq)\ (500\ mg\ KOH/g)\ (0.26)\ (1)}{(56.1\ g\ KOH/eq)\ (1000\ mg/g)} = 0.2586$$

$$T = 0.6804 + 0.2586 = 0.9390$$

Preparing activity:
Navy-OS
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3. DOCUMENT TITLE
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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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